

Plan and Steps to Transform (Clean the Data)

Before applying any data transformation steps, it's essential to thoroughly analyze the files to understand their structure, content, and potential inconsistencies.

STEP 1: ANALYZE "Dummy Data Customer Info.csv"

Identified Problems:

- Duplicate Customer Records

Example: customer_id = 100156 appears three times with different emails and birthdates.
Action: Decide on a deduplication rule, keep the most recent record.

- Date of Birth Format Inconsistency

Formats example used 93-06-14 (YY-MM-DD), 14/6/1993 (D/M/YYYY), 6/14/93 (M/D/YY)
1995/10/25 (YYYY/MM/DD)
Action: Standardize using a parser (e.g. convert all to YYYY-MM-DD).

Resolved Issues:

- Duplicate customer_id entries resolved by keeping the most recent birthdate.
- date_of_birth standardized to YYYY-MM-DD.
- Region names normalized (capitalized and stripped).
- No null values remain in required fields.

Python script: [clean_customer_info.py \(refer in the script folder\)](#)

STEP 2: ANALYZE "Dummy Data I TXN Part 1.csv and Dummy Data I TXN Part 2.csv"

Issues Identified in Part 1.csv

Field	Problem Description
CUTOMER_ID	Misspelled – should be CUSTOMER_ID
AMOUNT	String with commas, needs conversion to float
CURRENCY	All NaN, but actual currency is in ACCOUNT column (could be RWF,USD and EURO)
TRANSACTION_TYPE	Appears to be transaction direction (e.g. 'C' = Credit, 'D' = Debit)
COMMENT	Contains embedded info (e.g. "IDNO 024910", period, etc.), needs parsing

Issues Identified in Part 2.csv

Field	Problem Description
NAME, LAST NAME	Misaligned columns — these likely hold ACCOUNT, NAME, LASTNAME, CHANNEL
AMOUNT	In wrong column, includes description ("DISB LOAN JAN...") → needs extraction
ACCOUNT	Mostly missing

✓ Resolved Issues:

- Rename and Fix Column Names
- Standardize both files to:
 - CUSTOMER_ID
 - TRANSACTION_COMMENTS
 - AMOUNT
 - CURRENCY
 - ACCOUNT_NUMBER
 - NAME
 - LASTNAME
 - CHANNEL
- Normalize AMOUNT, Remove commas, extract the numeric part, and convert to float.
- Parse Extra Information, From TRANSACTION_COMMENTS, extract:
 - IDENTIFIER (e.g., IDNO 024910).
 - PERIOD (e.g., SEP-2015).

✓ Manual Data Fixes in Excel

Before applying the scripts, I resolved some simple modifications manually:

In "Dummy Data I TXN Part 1.csv"

- ✓ Corrected the typo CUTOMER_ID to CUSTOMER_ID.
- ✓ Moved the currency under the correct column.
- ✓ Moved the account number under the correct column.
- ✓ Adjusted the NAME and LASTNAME to their correct columns.
- ✓ Created a column called TRANSACTION_TYPE and assigned values based on "D" (Debit) and "C" (Credit).

In "Dummy Data I TXN Part 2.csv"

- ✓ Deleted data inside the COMMENT column from lines 4 to 36.
- ✓ Copied data from the AMOUNT column (lines 2 to 36) and pasted it into the COMMENT column.
- ✓ Copied data from column E (lines 2 to 41) and exchanged it with column F (lines 2 to 41).

Python script: **clean_transaction_data.py** (refer in the script folder)

Conclusion:

After applying all the outlined cleaning and transformation steps, the customer information will be fully standardized, with duplicate records resolved, date formats unified, and essential fields properly structured. The transaction data, originally fragmented across multiple files, will now be combined into a cohesive dataset with corrected column names, properly formatted amounts, and extracted key identifiers.

With these refined CSV files, the data will be ready for analysis—allowing for reliable insights into customer behavior, transaction trends, and financial patterns. Additionally, these cleaned datasets can be seamlessly loaded into a data warehouse, ensuring efficient storage, retrieval, and scalability for further processing or reporting.